VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA" Belagavi – 590018, Karnataka



A

MAJOR PROJECT REPORT

on

"VEERDRISHTI: COMBAT VISION SYSTEM"

Submitted in partial fulfillment for the award of the degree in

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Under the Guidance of

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2026 – 2027

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Major Project Report entitled **"VeerDrishti: Combat Vision System"** carried out by **Gaurav Chauhan [1BH23CS024] and Gaurav Singh [1BH23CS025]**, the bonafide student of **Bangalore Technological Institute**, **Bengaluru** in partial fulfilment for the award of **Bachelor of Engineering** in **Computer Science and Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2026-2027**. Thus, it is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report. The Major Project Report has been approved as it satisfies the academic requirements in respect of major project prescribed for the said degree.

Ms. Sapna N	**Mrs. Swaroopa K**
Asst. Prof., Project Guide	Asst. Prof., Project Coordinator
Dept. of CSE	Dept. of CSE
Dr. D Jerline Sheebha Anni	**Dr. H. S. Nanda**
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Name of Examiners	**Signature with date**
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the students of final year Computer Science and Engineering, **Bangalore Technological Institute, Bengaluru,** hereby declare that, the Major Project Report work **"VeerDrishti: Combat Vision System"** embodies the report of our research work carried out under the guidance of **Ms. Sapna N**, Asst. Prof., **Department of Computer Science and Engineering**, **Bangalore Technological Institute, Bengaluru** as partial fulfilment of requirements for the **Bachelor of Engineering Degree in Computer Science and Engineering** of **Visvesvaraya Technological University, Belagavi**, during the academic year 2026-2027.

We also declare that to the best of our knowledge and believe the work reported here does not form part of any other project on the basis of which a degree or award was conferred on an early occasion of this by any other.

Date: ……

Place: Bengaluru

Student Name	**USN**	**Signature**
GAURAV CHAUHAN	**1BH23CS024**	**……**
GAURAV SINGH	**1BH23CS025**	**……**

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ABSTRACT

In high-stress combat environments, operators must process a lot of visual information while dealing with constant positioning, movement, poor visibility, stress and mental fatigue. Continuous manual monitoring makes it difficult to spot threats, targets and points of interest quickly. It can slow decision-making leading to operational difficulties and casualties. VeerDrishti is an AI-powered computer vision framework that aims to provide real-time situational awareness and automated visual threat assessment. The system captures live video using a camera module and processes each frame using a on-device detection model. The detection pipeline does frame pre-processing, real-time inference and post-processing before it shows the detected objects with bounding boxes, labels and confidence scores on an augmented reality interface. The framework focuses on low-latency processing so that visual detection happens directly on edge hardware without an external computing infrastructure. So far tests have shown real-time operation at 18–25 frames per second with low latency on inexpensive low-end hardware under different conditions. Further evaluation on field conditions has to be done to assess accuracy, robustness and reliability in real-world operational environments.

VISION AND MISSION OF THE INSTITUTE

BANGALORE TECHNOLOGICAL INSTITUTE

VISION

To evolve as a Centre of Excellence in Academics, Research and Development with an impact on Technological advancement to the Society.

MISSION

M1- To provide young prospective engineers with an environment for higher education and research

M2–To mold students as responsible citizens with an attitude of using their knowledge for technological advancements

M3–To uplift the downtrodden and improve the quality of life of fellow human beings.

COMPUTER SCIENCE AND ENGINEERING

VISION

To impart the best in academia & research, empowering the students of Computer Science and Engineering by contributing their best for the society.

MISSION

M1 - To impart updated theoretical and practical knowledge in computer science and engineering by creating an outstanding learning environment.

M2 - To equip the students with wisdom in the discipline of computing and to apply their knowledge for benefiting the society.

M3 - To enhance research collaborations for upliftment of computing professionals.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1 - To showcase strong technical proficiency in computer science fundamentals, empowering the design, development and implementation of innovative solutions that address real world challenges.

PEO2 - To establish a solid foundation in the core concepts of computer science and engineering, while fostering a proactive attitude towards pursuing higher education and fulfilling industry demands.

PEO3 - To embody professional ethics, strong communication skills, and leadership qualities that faster multidisciplinary excellence and career growth.

PEO4 - To cultivate technical knowledge and research ability that inspire innovation, enabling the establishment of startups and fostering entrepreneurial success.

PEO5 - To tackle societal, environmental and global challenges through the design of sustainable and efficient computer science and engineering-based systems that improve overall quality of life.

PROGRAM OUTCOMES

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO1 - Demonstrate proficiency in computer education by applying principles of mathematics, science & engineering along with technical expertise.

PSO2 - Equip proficiency in computer science and engineering employing the principles of communication networks, database management systems, computing systems & standard software development practices to cater technical growth.

PSO3 - Develop proficiency in analytical skills, research & innovation contributing effectively as per the industry requirements by delivering effective socio - economic skills for the advancement in technology.